Cash transfers and parental time allocation: evidence from the 2021 extension of the child tax credit

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Abstract: In 2021, in response to the COVID-19 pandemic, the child tax credit in the US was extended as part of the American Rescue Plan. Using 2015 to 2022 data from the American Time Use Survey (ATUS) linked to the Current Population Survey (CPS), I estimate a difference-in-differences model to identify the impact the extension had on parental childcare time. I find that the policy extension reduced parental childcare time, consistent with households substituting for formal or paid childcare. However, the extension did not reduce the gap in domestic labor between mothers and fathers that was exacerbated during the pandemic. These results are robust to several specifications and sub-samples. My findings contribute to understanding how unconditional cash transfers affect household behavior and the gender heterogeneity in policy impact.

I. Introduction

Despite significant changes to women's role in the household and increases in female labor force participation across the last century, women are still responsible for the majority of childcare within a household in the United States (US). In households where both spouses are working full time, women have continued their pre-work force role of maintaining the housework and childcare, often referred to as the "second shift" (Hochschild, 1989). Women do not decrease their working time when they have children (Yavorsky et al., 2015) and as a result, lose out on income and face increased amounts of stress compared with their male partners (Mussida and Patimo, 2021). Moreover, the disproportionate amount of unpaid domestic work has a negative impact on women's mental health (Wang and Lu, 2023) and reinforces beliefs about gender roles (Croft, Schmader, Block, and Baron 2014) which precipitate the roles into future generations. With the onset of the COVID-19 pandemic and shut downs in the US in early 2020, childcare responsibilities increased significantly as schools and day cares were closed. Moreover, with the significant decrease in economic activity, many individuals were laid off and thus, households had less money for childcare, where it remained available. Post-pandemic research indicates that gender inequalities within domestic labor increased during pandemic lock downs (Obeng et al., 2022; Del Boca et al., 2020). As a result, women were more likely to experience anxiety, depression, or PTSD during the pandemic (Thibaut and ELNahas, 2023). Policymakers recognized the increased burden faced by families during the pandemic and passed the American Rescue Plan Act (ARPA) in 2021 which included an extension of the child tax credit (CTC).

A CTC is a common policy across many high income countries to credit families for expenses incurred as part of raising children. In 1997, the US introduced the CTC as a nonrefundable credit that households could claim in their tax filing. By 2015, the credit was partially refundable, $1,000 per child under 17, and households had to have at least $3,000 in earned income to be eligible with the credit phasing out at $110,000 per annum for married couples and $75,000 per annum for single individuals. In 2017, the CTC was increased from $1,000 to $2,000 per annum, the minimum income required to be eligible decreased to $2,500 per annum, and the maximum income thresholds where the credit phased out were increased to $400,000 per annum for married filing jointly taxpayers and $200,000 per annum for head of household or single taxpayers (Congressional Research Service, 2025). Introduction of this legislation gave eligible families more flexibility for work arrangements and childcare needs. Moreover, research has shown that government subsidies targeted at childcare increase female labor force participation (Averett et al., 1997), likely as a result of lower childcare burden.

In 2021, to address the challenges faced by families during the pandemic, the CTC was expanded for the 2021 tax year only, through the American Rescue Plan Act (ARPA). For children under 6, the credit was increased to $3,600 and for all other children under 18, the credit was increased to $3,000. Income eligibility for the extension was married couples making less than $150,000 per annum, head of household making less than $112,500 per annum, and singles making less than $75,000 per annum. Additionally, the expansion of the credit was

fully refundable so there was no minimum income required to receive the full amount of the credit. Households with income above these thresholds were eligible for the baseline credit as amended in 2017 (Congressional Research Service, 2025). Further, households could elect to receive half the extended credit as monthly advance payments in July to December 2021, with the remaining lump sum paid following tax filing in 2022 (Internal Revenue Service, 2021).

There is a significant amount of literature on the US CTC, mostly focused on the existence and impact of exclusion of low income households from receiving the credit in years other than 2021 (Goldin and Michelmore, 2022; Greenstein et al, 2018). For the 2021 CTC extension, research has shown that the expansion reduced child poverty in the US for 2021, improved child health outcomes (Aizer et al., 2024), and increased housing stability for low-income households (Pilkauskas et al., 2023). Some literature has looked to identify outcomes for parents such as an investigation of the potential effects of a permanent version of the 2021 extension that shows labor force participation of parents would decline (Bastian, 2024). Lebihan (2025) shows improved parental well-being and increased time spent with children during the monthly advanced payments in July 2021 through December 2021. However, some research indicates the effects on parents' psychological wellbeing is ambiguous (Gennetian and Gassman-Pines, 2023).

This paper builds on the parental outcome literature, specifically impacts on women's childcare time as a result of the 2021 CTC extension. Although the 2021 CTC extension was not focused on women specifically, as women generally have a higher burden in domestic labor, measuring the impact on mothers is imperative to identifying efficacy of policies aimed at families. I contribute to the growing literature on unconditional cash transfers (UCT) in the United States and gender heterogeneity in policy impacts. The COVID-19 pandemic showed that shocks affect men and women differently and thus, when considering responses to shocks, policymakers should consider outcomes for different genders. Moreover, as new policies such as universal day care and other UCTs are proposed, it is essential for policymakers to understand the role UCTs have in affecting women's domestic labor.

I use data for 2015 through 2022 from the American Time Use Survey (ATUS) and the Current Population Survey (CPS) to assess the impact the 2021 extension of the CTC had on women's hours spent on childcare using a difference-in-differences (DiD) model. I find some evidence that the 2021 CTC extension had an overall effect of reducing parental childcare time but did not have a statistically significant differential effect on mothers and fathers. A possible explanation is that the additional income from the CTC extension allowed households to substitute away from parent-provided childcare towards formal or paid childcare, without affecting the intra-household division of childcare between mothers and fathers.

II. Data

The ATUS is a cross-sectional survey conducted by the US Census Bureau (USCB) from a random representative subset of respondents to the CPS. The ATUS provides estimates for how Americans spend their time for a wide variety of activities such as childcare, housework, and volunteering. The estimates are based on interviews conducted by the USCB where respondents are asked questions about how they spent the last 24 hours. The USCB conducts interviews throughout the year (US Bureau of Labor Statistics, 2025). I have used data from the ATUS for 2015 through 2022 for respondents who have children. Since parents received portions of the CTC in July - December 2021 and 2022, I include data for both 2021 and 2022. As the 2021 rules did not extend into 2023 and 2024, I have not included these years.

To determine household composition and total household income, I have linked the ATUS with the CPS, which provides income and demographic data on the US population. I construct household income for each respondent to determine eligibility for the 2021 CTC extension. An individual as head of household (HOH) could receive the additional tax credit if they made less than $112,500 per annum and a couple, $150,000 per annum. A single individual, not HOH, could receive the credit if they made less than $75,000 per annum. I do not have access to individual's tax filing status from the CPS or ATUS to see if they did indeed file as HOH. However, to file as HOH, an individual must be unmarried, contribute more than 50% of expenses of the household, and have at least one dependent when filing their taxes. As my sample consists of only individuals with children, it is reasonable to assume that most of the unmarried individuals are filing as HOH. This assumption is supported by the fact that the majority of individuals filing under HOH are single parents (National Women's Law Center, 2017). Therefore, I have applied the HOH threshold for all unmarried individuals.

Table 1: Distribution of Households Eligible for the 2021 CTC Extension (Respondents interviewed July 2021 - December 2022)

	Count	**Percentage**
Ineligible	555	24.01%
Eligible	1,757	75.99%
Total	2,312	100%

Respondents interviewed after July 2021

Almost 76% of the sample was eligible for the 2021 CTC extension. I do not have access to a respondent's filing information and thus, cannot verify receipt of the CTC. However, as mentioned by Lebihan (2025), there are endogeneity issues with using CTC receipt as receipt is correlated with ability and motivation to apply which are affected by childcare time.

As the ATUS is pooled data, I have created a pseudo-eligibility for time prior to July 2021 to check for differences before the extension

between households that would have been eligible or ineligible had the extension occurred in an earlier period. Specifically, I apply the thresholds for the 2021 CTC extension to household income in each year prior in the sample to create a treated and control group. Using the distribution in Table 2, I will compare summary statistics for the treated and control households for the pre-July 2021 period.

Table 2: Distribution of Households Eligible for the 2021 CTC Extension for January 2015 - June 2021

	Count	**Percentage**
Ineligible	2,206	16.26%
Eligible	11,357	83.74%
Total	13,563	100%

The ATUS contains questions related to time spent with household children. I construct an individual's primary childcare time, where childcare was the sole activity being performed, from time reported for childcare activities listed in Appendix A. Additionally, the survey asks individuals about their secondary childcare time, time where the individual was caring for children while also performing another activity. I construct total childcare time from the addition of primary childcare time and secondary childcare time.

Table 3: Descriptive Statistics for Childcare Time of Sample (minutes per day) - Pre-July 2021 by Eligibility

	Count	Mean	Std. Dev.	Min	Max
Ineligible for 2021 CTC Extension					
Time spent caring for household children as primary activity	2,206	82.23	104.54	0	780
Time spent caring for household children as secondary activity	2,206	288.23	288.92	0	1260
Total time spent caring for household children	2,206	370.46	321.67	0	1290
Eligible for 2021 CTC Extension					
Time spent caring for household children as primary activity	11,357	67.60	102.06	0	945
Time spent caring for household children as secondary activity	11,357	264.58	282.12	0	1185
Total time spent caring for household children	11,357	332.18	313.87	0	1230
Observations	13,563				
Difference in Mean (Ineligible − Eligible - Total CC Time)	38.28				
Standard Error of Difference	7.33				
t-statistic	5.22				
p-value (two-sided)	0.00				

As can be seen in Table 3, for households that were ineligible, respondents had on average, about 38 minutes of extra childcare per day, a

statistically significant difference. Under the 2021 CTC extension rules, the households who are ineligible are wealthier, likely have more flexibility in work schedules, and spend more time on childcare. Additionally, single parents are more likely to belong to the eligible group and may face more pressure to work multiple jobs or less flexible hours, reducing their childcare time.

To consider the differing effect the CTC extension had on men versus women, I use indication from the CPS for the sex of each respondent to the ATUS.

Table 4: Gender Distribution - Pre-July 2021 by Eligibility

	Ineligible (N = 2,206)	**Eligible** (N = 11,357)
Male	1,385 (62.78%)	5,816 (51.21%)
Female	821 (37.22%)	5,541 (48.79%)
Total	2,206 (100%)	11,357 (100%)
Difference in means (Female share)	-0.1157	
t-statistic	-10.00	
p-value (two-sided)	0.00	

As can be seen in Table 4, the eligible group is more evenly split by sex than the ineligible group, a statistically significant difference. This could be driven by differentials in the earnings of men and woman or family structures. For example, married households are more likely to be eligible to receive the CTC (Goldin and Michelmore, 2022). Additionally, Brehm and Malkova (2023) indicate that single mothers saw significant gains in eligibility by 2017 and surpassed single fathers and narrowed the gap to married couples. These factors likely contribute to the varied split.

For control variables, I include dummy variables for individuals with a partner, household inclusion and employment of the partner, if adult relatives live with them, and education level of the respondent. Additional controls are included for weekly work hours, number of children, and age of the respondent's youngest child. Summary statistics for control variables are included in Appendix B. I also include fixed effects for the state the individual lives in to capture time-invariant effects across states and fixed effects for the year of the cross-section to absorb any time-related shocks. Further, I include fixed effects for month and weekday that the ATUS interview was conducted on to capture any differences in childcare time based on time of the year or weekend versus school days.

Additionally, since I use data beginning in 2015 and there was a change to the CTC in 2017, I include a control for eligibility in terms of the 2017 expansion. The income thresholds for the 2017 expansion included more households but the minimum income threshold was still in place under these rules. This floor component is a significant driver of ineligibility, accounting for the driver of ineligibility for 80% of children who do not qualify (Goldin and Michelmore, 2022).

Table 5: 2017 CTC Eligibility - Pre-July 2021 by 2021 Eligibility

	Ineligible in 2021 (N = 2,206)	**Eligible in 2021** (N = 11,357)
Not Eligible for CTC in 2017	636 (28.83%)	4,137 (36.43%)
Eligible for CTC in 2017	1,570 (71.17%)	7,220 (63.57%)
Total	2,206 (100%)	11,357 (100%)

As can be seen in Table 5, a significant number of respondents that would have been eligible under 2021 terms, were ineligible under 2017 terms.

III. Empirical Approach

In order to estimate the effect the 2021 CTC extension had on women's childcare time, I use a DiD model specified as follows:

$$\ln(ChildcareTime)_{it} = \beta_1 Eligible_i + \beta_2 Post2021_t + \beta_3 CTCit + \beta_4 Female_i + \gamma X_{it} + +\lambda_y + \eta_m + \pi_d + \alpha_s + \epsilon_{its} \quad (1)$$

The outcome in my specification is the natural log of total childcare time, where childcare time is reported in minutes. Since some respondents report zero childcare time, I take the $Log(ChildcareTime + 1)$ which allows the log specification without dropping observations while preserving zeros in the model.

Eligibility or pseudo eligibility of a respondent is given by $Eligible_i$ which indicates a respondents eligibility under the 2021 CTC extension terms for any year. Additionally, $Post2021_t$ indicates that the interview is after July 2021 and thus, that the extension was in effect. The variable CTC_{it} captures eligibility for the 2021 CTC extension and that it is after July 2021. Thus, this variable represents the treatment and is equivalent to $Eligible_i \times Post2021_t$.

Controls in the model, as discussed in the data section, are given by X_{it} and year, month, day, and state fixed effects are given by λ_y, η_m, π_d, and α_s, respectively. Lastly, the standard errors clustered at the state level are given by ϵ_{ist}. I cluster the standard errors at the state level as there are state-specific characteristics that will influence households income, childcare constraints, and demographics and thus, standard errors are likely correlated within states. Additionally, the standard error terms within a state are likely not independent with time. Hence, estimates are robust to heteroskedasticity and I limit overstating the statistical significance of the estimates. Since fixed effects for time are included in the model, the concise specification drops $Post2021_t$ as follows:

$$\ln(ChildcareTime)_{it} = \beta_1 Eligible_i + \beta_3 CTCit + \beta_4 Female_i + \gamma X_{it} + \lambda_y + \eta_m + \pi_d + \alpha_s + \epsilon_{its} \quad (2)$$

The effect of the 2021 CTC extension alone is given by $e^{\beta_3} - 1$ in terms of percent change of childcare time.

To consider gender heterogeneity of the treatment, I include an interaction term between the treatment and the individual's sex. Further, I control for differences between women who meet eligibility requirements given by $Female_i \times Eligible_i$ and shocks that affected only women after 2021 given by $Female_i \times Post2021_t$.

$$\ln(ChildcareTime)_{it} = \beta_1 Eligible_i + \beta_2 Post2021_t + \beta_3 CTCit + \beta_4 Female_i + \beta_5 Female_i \times Eligble_i$$

$$+\beta_6 Female_i \times Post2021_t + \beta_7 Female_i \times CTCit + \gamma X_{it} + +\lambda_y + \eta_m + \pi_d + \alpha_s + \epsilon_{its} \qquad (3)$$

The effect for women who were eligible for the extension is given by $e^{\beta_3+\beta_5} - 1$ in terms of percent change of childcare time.

The counterfactual in my model is the households which were not eligible for the 2021 CTC extension, which are households with higher income. My identification strategy relies on the assumption that the two groups, treated and counterfactual, were experiencing a common trend prior to the CTC extension. The following chart plots the trends of both groups leading up to July 2021 where the counterfactual is given by the dashed line and the households who were eligible is given by the solid line. As mentioned above, eligibility prior to July 2021 is based on applying the same terms for the 2021 CTC extension to household income of respondents in each time period.

Figure 1: Childcare Time of Ineligible versus Eligible of 2021 CTC Extension Households (January 2015 – June 2021) - Monthly Average

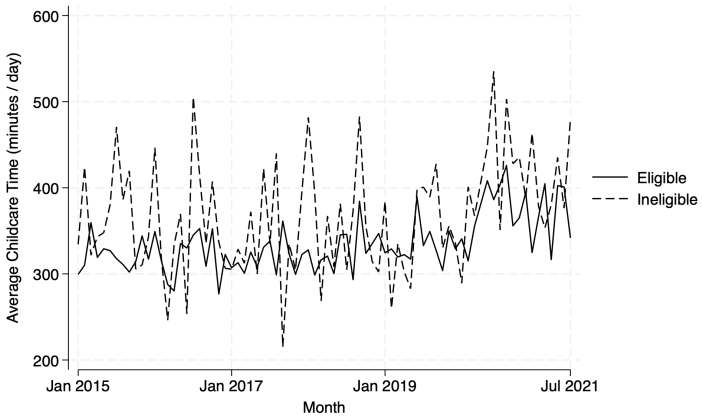
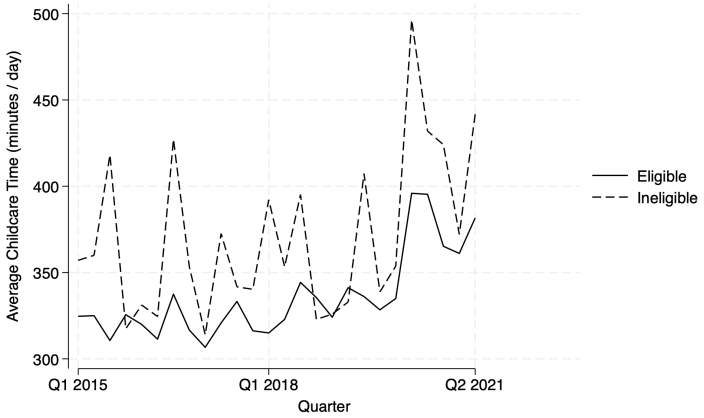


Figure 2: Childcare Time of Ineligible versus Eligible of 2021 CTC Extension Households (January 2015 – June 2021) - Quarterly Average



Generally, across the months, both the treated and control group have volatile trends in childcare time. Visually, it appears as though ineligible or wealthier households have higher fluctuations in their childcare time. Particularly, we see some spikes in summer months which may represent the flexibility wealthier households have to spend with their children in non-school months. This plot alone does not verify or invalidate the parallel trends assumption. To formally test for parallel trends, I conduct a placebo test by interacting pseudo-eligibility with the period prior to July 2021, including controls and fixed effects mentioned in the data section. Assuming that the parallel trend assumption holds, the treatment indicator should be near zero and not statistically different from zero. Otherwise, it implies that eligibility under 2021 terms was causal prior to the extension of the CTC in July 2021.

$$\ln(ChildcareTime)_{it} = \beta_1 Eligible_i + \beta_2 Eligible_i \times Time_{PreJuly2021} + \gamma X_{it} + \lambda_y + \eta_m + \pi_d + \alpha_s + \epsilon_{its} \quad (4)$$

In equation (4), $Time_{PreJuly2021}$ represents the time period prior to July 2021. Results from the regression are as follows:

Table 6: Placebo Test for Parallel Pre-Treatment Trends

VARIABLES	(1) Placebo Test
Eligible	-0.00133
	(0.0686)
Pre-July 2021 Period	0.00858
	(0.00898)
Eligible × Pre-July 2021 Period	0.00110
	(0.00161)
Observations	13,566
R-squared	0.381

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1
Fixed Effects and Controls included in model but omitted from table

Indeed, the coefficient for $Eligible_i \times Time_{Pre2021}$ is near zero and not statistically significant. Additionally, I have interacted treatment with years and quarters and find no statistically significant coefficients, where results are included in Appendix C. Therefore, the parallel trends assumption holds.

IV. Results

Results are presented in Table 7 across four specifications. Column (1) reports a simple DID model with state and time fixed effects only. Column (2) adds controls for demographics, household composition, and employment characteristics. The coefficient on CTC_{it} is somewhat statistically significant in Model 2, implying an associated 14% reduction in childcare time for households eligible for the extension. This estimate reflects the average effect across both mothers and fathers.

Column (3) includes gender interactions to assess the differential impact of the 2021 CTC extension on mothers and fathers. Under this specification, the coefficient on CTC_{it} captures the effect for fathers and the coefficient on $CTC_{it} \times Female_i$ captures the effect for mothers. Both gender interaction terms are statistically insignificant, implying no evidence of gender heterogeneity for the 2021 CTC extension's impact on childcare time. Column (4) further controls for eligibility for the 2017 expansion to address any potential overlapping effect of earlier policy changes. The coefficients for the 2021 CTC extension remain negative but also not statistically different from zero.

Overall, there is some evidence that the 2021 CTC extension reduced childcare time of parents but no evidence that it affected mothers and fathers differently.

Table 7: Effects of the 2021 CTC Extension on Childcare Time

VARIABLES	(1) Log(Childcare Time)	(2) Log(Childcare Time)	(3) Log(Childcare Time)	(4) Log(Childcare Time)
Eligible	-0.265***	0.0492	0.0143	0.0139
	(0.0459)	(0.0390)	(0.0458)	(0.0457)
CTC	-0.263***	-0.151*	-0.121	-0.123
	(0.0968)	(0.0830)	(0.108)	(0.109)
Female		0.137***	0.0475	0.0470
		(0.0434)	(0.0781)	(0.0779)
Female×Post2021			0.138	0.138
			(0.146)	(0.146)
Female × Eligible			0.104	0.105
			(0.0968)	(0.0967)
Female × CTC			-0.148	-0.147
			(0.177)	(0.177)
Eligible for 2017 CTC				0.513
				(0.434)
Observations	15,875	15,875	15,875	15,875
R-squared	0.015	0.379	0.379	0.379
State Fixed Effects	Yes	Yes	Yes	Yes
Time Fixed Effects	Yes	Yes	Yes	Yes
Demographic Controls	No	Yes	Yes	Yes
HH Composition Controls	No	Yes	Yes	Yes
Employment Controls	No	Yes	Yes	Yes
2017 CTC Eligibility	No	No	No	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Robustness

Table 8 presents two robustness checks for including childcare costs (columns 1 and 2) and COVID-19 reopening intensity across states in the model (columns 3 and 4). Childcare costs are derived from yearly state-level cost data from the National Database of Childcare Prices. As costs can change across years, the costs are not inherently captured in the state fixed effects and may influence parent's childcare time allocation. I use the Oxford COVID-19 Government Response Tracker Stringency Index to control for differing reopening intensity across states which affected household access to formal childcare.

Under these specifications, the overall effect of the treatment becomes statistically insignificant but similar in magnitude. Under the gender heterogeneous specification, the effect on mothers versus fathers remains insignificant and also similar in magnitude. Hence, the

baseline model is not sensitive to these additional controls. Moreover, the effect of childcare costs and reopening intensity are very close to zero and not statistically significant, suggesting that neither meaningfully influence childcare time in this sample.

Table 8: Robustness Check for Childcare Cost and COVID-19 Reopening Intensity

	(1)	(2)	(3)	(4)
VARIABLES	Log(Childcare Time)	Log(Childcare Time)	Log(Childcare Time)	Log(Childcare Time)
Eligible	0.0895**	0.0649	0.0476	0.0322
	(0.0389)	(0.0423)	(0.0382)	(0.0440)
CTC	-0.118	-0.0717	-0.141	-0.0994
	(0.0995)	(0.121)	(0.0889)	(0.112)
Female	0.137***	0.0816	0.137***	0.0380
	(0.0461)	(0.0754)	(0.0434)	(0.0741)
Female × Time		0.0978		0.181
		(0.153)		(0.154)
Female × Eligible		0.0712		0.111
		(0.101)		(0.0927)
Female × CTC		-0.155		-0.157
		(0.178)		(0.181)
State Childcare Cost	-2.53e-05	-4.80e-05		
	(0.00175)	(0.00176)		
Stringency Index			0.000783	0.000849
			(0.00197)	(0.00198)
Observations	14,279	14,279	15,875	15,875
R-squared	0.384	0.384	0.379	0.386

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1
Fixed Effects and Controls included in model but omitted from table

I have performed an additional robustness check to address potential differences between the 2015-2018 period and 2019-2022. Because the 2021 CTC extension was implemented in response to the COVID-19 emergency, the 2015-2018 period may reflect too different of an environment. Thus, as a check, I restrict the sample to 2019 to 2022 which spans just before the pandemic and the introduction of the extension. The 2019-2022 sample offers a more direct pre-treatment comparison that does not have confounding from the earlier 2017 policy change. However, this exclusion reduces the sample significantly and thus, reduces the statistical power.

As can be seen in Table 9, the overall effect of the treatment becomes slightly more negative and is statistically significant, implying an associated 16.7% reduction in childcare time for households eligible for the 2021 CTC extension. However, again we see similar results to the baseline model for the difference in effect for mothers and fathers. These results support the argument that the main findings are not

significantly affected by the earlier period, 2015-2018.

Table 9: Robustness Check for 2019 - 2022

VARIABLES	(1) Log(Childcare Time)	(2) Log(Childcare Time)
Eligible	0.102	0.114
	(0.0763)	(0.0929)
CTC	-0.183**	-0.150
	(0.0887)	(0.126)
Female	0.130**	0.0432
	(0.0611)	(0.130)
Female × Post2021		0.210
		(0.159)
Female × Eligible		0.0783
		(0.141)
Female × CTC		-0.146
		(0.189)
Observations	6,793	6,793
R-squared	0.392	0.400

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1
Fixed Effects and Controls included in model but omitted from table

The final robustness check removes all 2022 responses from the sample. Households could receive half of the refund as monthly advance payments between July and December 2021, with the remaining lump sum received through the normal tax filing process in Spring 2022. In excluding 2022, I identify any difference in effect from receiving the monthly payments versus the lump sum.

Again, the results are similar to the baseline with a reduction associated with the treatment on an overall basis and no statistically significant difference in the effect on mothers versus fathers.

Table 10: Robustness Check for Monthly Advance Payments

VARIABLES	(1) Log(Childcare Time)	(2) Log(Childcare Time)
Eligible	0.0465	0.0305
	(0.0395)	(0.0444)
CTC	-0.137	-0.138
	(0.0937)	(0.121)
female	0.138***	0.0438
	(0.0419)	(0.0728)
Female × Post2021		0.203
		(0.281)
Female × Eligible		0.106
		(0.0914)
Female × CTC		-0.125
		(0.274)
Observations	14,402	14,402
R-squared	0.380	0.386

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1
Fixed Effects and Controls included in model but omitted from table

Overall, across specifications, the empirical results somewhat indicate that the 2021 CTC extension reduced household's overall childcare time on an aggregate basis by but did not differentially affect mothers or fathers' childcare time. Hence, these findings indicate that the temporary policy extension may have allowed household's to substitute towards paid or formal childcare by improving financial stability. However, this is not a gender specific reallocation but rather both parents reduced care as a result of being eligible for the 2021 CTC extension.

V. Conclusion

This paper offers empirical evidence on the effect the 2021 CTC extension had on parents' childcare burden. I show that the policy allowed parents to reduce their childcare time likely by switching towards more formal childcare offerings. These results contradict previous findings suggesting that the 2021 CTC extension increased parents' childcare time and aligns more with results that showed ambiguous effects. Further, the paper considers heterogeneous effects and upholds limited effects across genders. Lastly, the results contribute to ongoing assessment of UCTs which have also found mixed results on various components of childcare (Dacass et al., 2024; Krause et al., 2025).

The evidence in this paper suggests that cash transfers affect how households allocate their time but only on an aggregate basis. Specif-

ically, the 2021 CTC extension did not reduce childcare gaps across gender that were widened during COVID-19 shut downs. A possible explanation is that the gaps exacerbated by COVID-19 are deeply structural and not easily alleviated through a cash transfer. Moreover, domestic labor patterns take a long time to adjust and would likely benefit more from a long-term CTC extension. Additionally, the CTC does not cater to mothers specifically and thus, to address domestic labor gaps, other policies should be considered in addition to the CTC such as paid parental leave or subsidized child care.

Despite the findings, there are limitations to my results. First and most significantly, the the overall effect of the 2021 CTC extension is only moderately statistically significant and loses significance with some robustness checks. Thus, the evidence needs further support from future research to have external validity.

Second, there is the potential existence of measurement error as ATUS relies on self reported childcare time and only captures a respondent's time diary on a single day. Further, I impute eligibility for the 2021 CTC extension using CPS data and cannot observe filing status or CTC receipt. Therefore, there could be some misclassification of treatment status. However, with CTC receipt, there is endogeneity and thus, eligibility, which I use, is a more appropriate treatment indicator for causal identification. Although eligibility is a common instrument for receipt, I cannot test its strength.

Third, the timing of this study is during the COVID-19 recovery period and there could be residual unobservable variables such as work from home policies and household preferences. The included robustness check for COVID-19 reopening policies across states deals with some of these but work from home arrangements and household preferences are not observable for the sample. Fourth, the 2021 CTC extension was a temporary policy change and thus, these results do not capture long-term behavior of households. Moreover, since this was a short-term policy, there is limited interpretation for gender heterogeneity as gender norms and roles can be very slow to change. Hence, these results do not identify any of these gender norms, household bargaining, or labor force flexibility of households which could be contributing to domestic labor gaps. Thus, with these results, I cannot comment on any reduction or further widening of the gap.

Future work should try to incorporate richer administrative data or longitudinal data and compare outcomes from the 2021 CTC extension to longer-term policy expansions or non-COVID-19 related policy expansions. Additionally, one could look at the effect of policies targeted at mothers specifically or further outcomes that occur as a result of reducing maternal childcare time. For example, future research could assess the impact of reduced childcare time for mothers on persistent working hours, promotions, or job mobility. Moreover, one could look at payment routing to mothers only and the effect on caregiving behavior. Considering gendered outcomes addresses the differing constraints, incentives, and social expectations faced by different genders and can inform policymakers of the impact of shifting from gender-neutral to gender-aware policy frameworks.

VI. References

Aizer, A., Lleras-Muney, A., & Michelmore, K. (2023). The effects of the 2021 Child Tax Credit on child developmental outcomes. *The ANNALS of the American Academy of Political and Social Science*, 710(1), 172–189.

Averett, S. L., Peters, H. E., & Waldman, D. M. (1997). Tax credits, labor supply, and child care. *Review of Economics and Statistics*, 79(1), 125–135.

Bastian, J. (2024). How would a permanent 2021 Child Tax Credit expansion affect poverty and employment? *National Tax Journal*, 77(2), 263–311.

Brehm, M. E., & Malkova, O. (2023). The child tax credit over time by family type: Benefit eligibility and poverty. *National Tax Journal*, 76(3), 707–741.

Congressional Research Service. (2025, March 19). *The Child Tax Credit: Legislative History* (CRS Report No. R45124). https://www.congress.go product/R45124

Croft, A., Schmader, T., Block, K., & Baron, A. S. (2014). The second shift reflected in the second generation: Do parents' gender roles at home predict children's aspirations? *Psychological Science*, 25(7), 1418–1428.

Dacass, T., Dilden, E., & Jiao, Y. (2024). How Does the Child Tax Credit Change the Time Allocation of Parents? Evidence from American Time Use Data. Evidence from American Time Use Data (October 25, 2024).

Del Boca, D., Oggero, N., Profeta, P., & Rossi, M. (2020). Women's and men's work, housework and childcare, before and during COVID-19. *Review of Economics of the Household*, 18(4), 1001–1017.

Gennetian, L. A., & Gassman-Pines, A. (2023). The effects of the 2021 Child Tax Credit on parents' psychological well-being. *The ANNALS of the American Academy of Political and Social Science*, 710(1), 124–140.

Goldin, J., & Michelmore, K. (2022). Who benefits from the child tax credit?. National Tax Journal, 75(1), 123-147.

Greenstein, R., Maag, E., Huang, C. C., Horton, E., & Cho, C. (2018). Improving the child tax credit for very low-income families. Washington, DC: US Partnership on Mobility from Poverty.

Hale, T., Angrist, N., Goldszmidt, R., Kira, B., Petherick, A., Phillips, T., Webster, S., Cameron-Blake, E., Hallas, L., Majumdar, S., & Tatlow, H. (2020–2022). Oxford COVID-19 Government Response Tracker (OxCGRT): Stringency Index [Data set]. Blavatnik School of Government, University of Oxford. https://www.bsg.ox.ac.uk/covidtracker

Hochschild, A. R. (1989). *The Second Shift: Working Parents and the Revolution at Home*. Penguin Books.

Internal Revenue Service. (2021). 2021 child tax credit and advance child tax credit payments: Topic A — General information. https://www.irs.gov/credits-deductions/2021-child-tax-credit-and-advance-child-tax-credit-payments-topic-a-general-information

Krause, P. K., Rhodes, E., Miller, S., Bartik, A. W., Broockman, D. E., & Vivalt, E. (2025). The Impact of Unconditional Cash Transfers on Parenting and Children (No. w34040). National Bureau of Economic Research.

Lebihan, L. (2025). Effects of the 2021 expanded child tax credit on parents' well-being and time use. *Economics & Human Biology*, 101533.

Mussida, C., & Patimo, R. (2021). Women's family care responsibilities, employment and health: A tale of two countries. *Journal of Family and Economic Issues*, 42(3), 489–507.

National Women's Law Center. (2017, July). *Eliminating the head of household filing status would hurt women* [Fact sheet]. https://nwlc.org/wp-content/uploads/2017/08/Eliminating-the-Head-of-Household-Filing-Status-Would-Hurt-Women.pdf

Obeng, C., Slaughter, M., & Obeng-Gyasi, E. (2022). Childcare issues and the pandemic: Working women's experiences in the face of COVID-19. *Societies*, 12(4), 103.

Pilkauskas, N. V., Michelmore, K., & Kovski, N. (2023). The 2021 Child Tax Credit, the living arrangements and housing affordability of families with low incomes (No. w31339). *National Bureau of Economic Research*.

Thibaut, F., & ELNahas, G. (2023). Women's mental health and lessons learnt from the COVID-19 pandemic. *Psychiatric Clinics of North America*, 46(3), 415–426.

U.S. Bureau of Labor Statistics. (2025, August 15). *American Time Use Survey: Concepts*. In *Handbook of Methods*. Retrieved from https://www.bls.gov/opub/hom/atus/concepts.htm

U.S. Bureau of Labor Statistics. (2015–2022). American Time Use Survey (ATUS) [Data set]. https://www.bls.gov/tus/

U.S. Census Bureau & U.S. Bureau of Labor Statistics. (2015–2022). Current Population Survey (CPS) [Data set]. https://www.census.gov/prograr surveys/cps/

U.S. Department of Labor, Women's Bureau. (2023). National Database of Childcare Prices (NDCP) [Data set]. https://www.dol.gov/agencies/wb/

Wang, S., & Lu, Z. (2023). Is paid inflexible work better than unpaid housework for women's mental health? The moderating role of parenthood. *Applied Research in Quality of Life*, 18(1), 393–409.

Yavorsky, J. E., Kamp Dush, C. M., & Schoppe-Sullivan, S. J. (2015). The production of inequality: The gender division of labor across the transition to parenthood. *Journal of Marriage and Family*, 77(3), 662–679.

VII. Appendices

Appendix A

Table 11: Time Categories Considered for Primary Childcare from ATUS (All Household Children)

Category	Mean	Std. Dev.	Min	Max
Physical care for children	11.8	36.8	0.0	897.0
Reading to children	1.2	7.0	0.0	210.0
Playing with children	8.6	38.8	0.0	750.0
Arts & crafts with children	0.1	2.9	0.0	210.0
Sports with children	0.4	5.5	0.0	240.0
Talking / listening to children	1.2	9.5	0.0	300.0
Organizing and planning for children	0.4	5.0	0.0	420.0
Looking after children as primary activity	2.0	19.9	0.0	900.0
Attending children events	2.5	22.8	0.0	810.0
Waiting for / with children	0.5	5.2	0.0	300.0
Picking up / dropping off children	1.2	5.2	0.0	147.0
Caring and helping for children	0.2	8.2	0.0	960.0
Homework with children	1.5	12.3	0.0	595.0
Meetings and school conferences	0.1	2.9	0.0	230.0
Homeschooling of children	0.2	6.5	0.0	480.0
Providing medical care	0.2	6.9	0.0	560.0
Obtaining medical care for children	0.2	5.3	0.0	350.0
Waiting associated with children's health	0.1	2.2	0.0	160.0
Waiting associated with children's education	0.0	0.6	0.0	90.0
Travel related to children's education	0.1	2.2	0.0	240.0
Travel related to children's health	0.1	2.6	0.0	180.0

Appendix B

Table 12: Household Composition - Pre-July 2021 by Eligibility

	Ineligible (N = 2,206)	**Eligible** (N = 11,357)
Has a Partner		
No	140 (6.35%)	2,875 (25.34%)
Yes	2,066 (93.65%)	8,482 (74.66%)
Partner in Household		
No	206 (9.34%)	3,249 (28.62%)
Yes	2,000 (90.66%)	8,108 (71.38%)
Partner Employed		
No	253 (11.47%)	4,749 (41.82%)
Yes	1,953 (88.53%)	6,608 (58.18%)
Has Adult Relatives in Household		
No	2,146 (97.28%)	10,619 (93.50%)
Yes	60 (2.72%)	738 (6.50%)
Total	2,206 (100%)	11,357 (100%)

Table 13: Education Level - Pre-July 2021 by Eligibility

	Ineligible (N = 2,206)	**Eligible** (N = 11,357)
Education Level		
High School or Less	144 (6.53%)	3,223 (28.38%)
College or Some College	1,016 (46.06%)	6,254 (55.07%)
Graduate Degree	1,046 (47.42%)	1,880 (16.55%)
Total	2,206 (100%)	11,357 (100%)

Table 14: Descriptive Statistics for Number of Children, Age of Youngest Child of Respondent, and Weekly Working Hours of Respondent - Pre-July 2021 by Eligibility

	Count	Mean	Std. Dev	Min	Max
Ineligible					
Number of Children	2,206	1.77	0.75	1.00	8.00
Age of Youngest Child	2,206	7.30	5.18	0.00	17.00
Work Hours per week	2,206	43.08	11.11	1.00	100.00
Eligible					
Number of Children	11,357	1.80	0.90	1.00	8.00
Age of Youngest Child	11,357	7.32	5.10	0.00	17.00
Work Hours per Week	11,357	39.73	11.98	0.00	100.00

Appendix C

Table 15: Placebo Parallel Trends Test by Year

VARIABLES	(1) Placebo - Yearly
Eligible	-0.0853
	(0.138)
2015 × Eligible	0.0921
	(0.164)
2016 × Eligible	-0.00521
	(0.177)
2017 × Eligible	0.240
	(0.186)
2018 × Eligible	0.0746
	(0.174)
2019 × Eligible	0.228
	(0.176)
2020 × Eligible	0.173
	(0.179)
Observations	13,695
R-squared	0.380

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1
Fixed Effects and Controls included in model but omitted from table

Table 16: Placebo Parallel Trends Test by Quarter

VARIABLES	(1) Placebo - Quarterly
Eligible	-0.271
	(0.425)
Q1 2015 × Eligible	0.398
	(0.473)
Q2 2015 × Eligible	0.284
	(0.511)
Q3 2015 × Eligible	-0.172
	(0.461)
Q4 2015 × Eligible	0.527
	(0.492)
Q1 2016 × Eligible	0.0195
	(0.484)
Q2 2016 × Eligible	0.316
	(0.489)
Q3 2016 × Eligible	0.0796
	(0.492)
Q4 2016 × Eligible	0.313
	(0.522)
Q1 2017 × Eligible	0.373
	(0.456)
Q2 2017 × Eligible	0.251
	(0.552)
Q3 2017 × Eligible	0.621
	(0.457)
Q4 2017 × Eligible	0.417
	(0.467)
Q1 2018 × Eligible	0.101
	(0.429)
Q2 2018 × Eligible	0.259
	(0.491)
Q3 2018 × Eligible	0.251
	(0.505)
Q4 2018 × Eligible	0.442
	(0.399)
Q1 2019 × Eligible	0.650
	(0.520)
Q2 2019 × Eligible	0.318
	(0.515)
Q3 2019 × Eligible	0.209
	(0.524)
Q4 2019 × Eligible	0.465
	(0.530)
Q1 2020 × Eligible	0.370
	(0.489)
Q2 2020 × Eligible	0.143
	(0.450)
Q3 2020 × Eligible	0.525
	(0.537)
Q4 2020 × Eligible	0.337
	(0.453)
Q1 2021 × Eligible	0.461
	(0.483)
Q2 2021 × Eligible	-0.0997
	(0.408)
Observations	13,695
R-squared	0.381

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Fixed Effects and Controls included in model but omitted from table